AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 11 · STUDENT LESSON

Training and Testing

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Define the claim. Keep groups together. Audit the evidence.

Distinguish frames from trips and scans from books before interpreting any held-out score.

Keep one habit: New-example claim → assignment and counts → overlap/completeness audit → independent repair.

MAIN LESSON · SECTION 1

What kind of new example should the test represent?

> Choose the evaluation claim before assigning related records.

Your learning goals

I can choose a split unit that matches the intended **prediction** claim.

I can keep related examples together and audit **group** overlap.

I can record split roles counts and limits before model evaluation.

Recall why agreement does not establish correct labels

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Why does agreement between two annotators not prove their labels are correct?

Your split-audit outcome

By the end, I can split related records by **group** when evaluation concerns new **group**s.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Human Classifier Simulation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Operational Labels. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Loops and Stopping. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 11: **Training** and Testing

The **test set** estimates performance on **unseen** cases.



THINKER MISSION

Create a fair split before anyone sees the mystery cards.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Keep all frames from one trip in one role

> This example concerns new trips on a familiar route.

The claim and group definition

A model may seem successful when its test examples closely resemble examples it already learned from. Imagine several pictures taken seconds apart during one robot trip.

They have different picture IDs but share a route, lighting and objects. This week you will assign entire trips to separate sets and explain what the resulting evaluation could measure. No model is fitted in this lesson.

Our invented twelve-row dataset contains frame descriptions from six trips. Each row is one frame; the **group** is its trip ID. The proposed claim concerns new trips on the same practice route under similar conditions.

We therefore keep all frames from a trip in one set. To claim success on new routes, **group**ing only by trip would not be enough.

The complete twelve-frame assignment

Trip group	Frame IDs	Assigned role
A	A1 A2 A3	Training
B	B1 B2	Training
C	C1	Training
D	D1 D2	Validation

Trip group	Frame IDs	Assigned role
E	E1 E2	Test
F	F1 F2	Test

What this small invented dataset can establish

This small fictional assignment is for learning split design, not **evidence** of real performance. Each frame belongs to one **group** and one set.

Store frame and trip IDs as metadata for auditing; do not assume an ID is an appropriate **prediction input**.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use **training** and testing to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

Audit group overlap with a complete Python script

> A clean group list is one check, not proof that the dataset is sound.

Give training, validation and test different jobs

Training data fits the model. Validation data helps compare choices such as candidate rules. Keep the **test set** for a final check after those choices are fixed.

A learner who repeatedly changes the model after seeing test results has used that set for development; its score is no longer an untouched final check.

Predict, then run the supplied intersection audit

```
train_groups = {"A", "B", "C"}
valid_groups = {"D"}
```

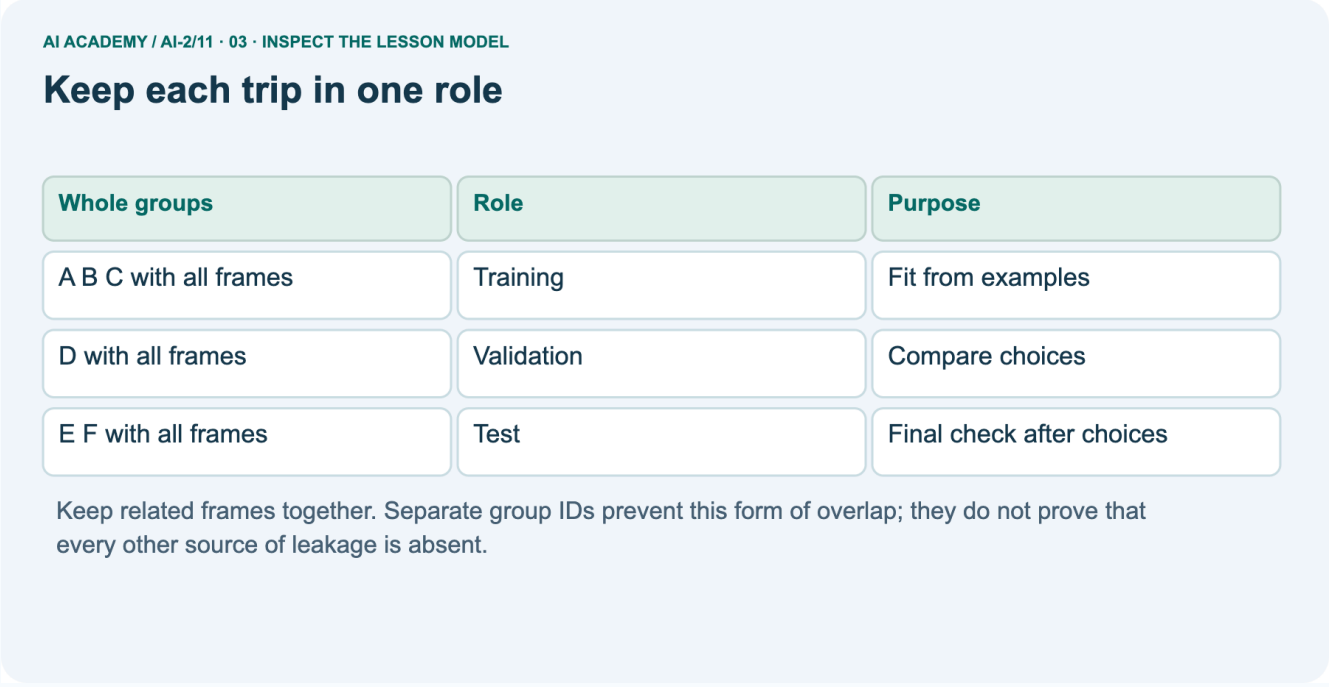
```
test_groups = {"E", "F"}
shared = ((train_groups & valid_groups)
          | (train_groups & test_groups)
          | (valid_groups & test_groups))
print(sorted(shared))
print(len(train_groups), len(valid_groups),
      len(test_groups))
```

Interpret empty overlap and three group counts

The first line of **output** is []: no trip ID is shared between any pair of sets. The second is 3 1 2: **training**, validation and test contain three, one and two trip **group**s. The symbol & finds shared values and | combines sets.

These are **group** counts, not frame counts.

The three roles in the diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

State what the script cannot detect

This audit assumes the lists correctly describe every frame assignment. It cannot find an omitted frame, a wrong trip ID, a copied image with a different ID or a feature containing the answer.

Check the assignment table and source records too.

Explain a deciding step

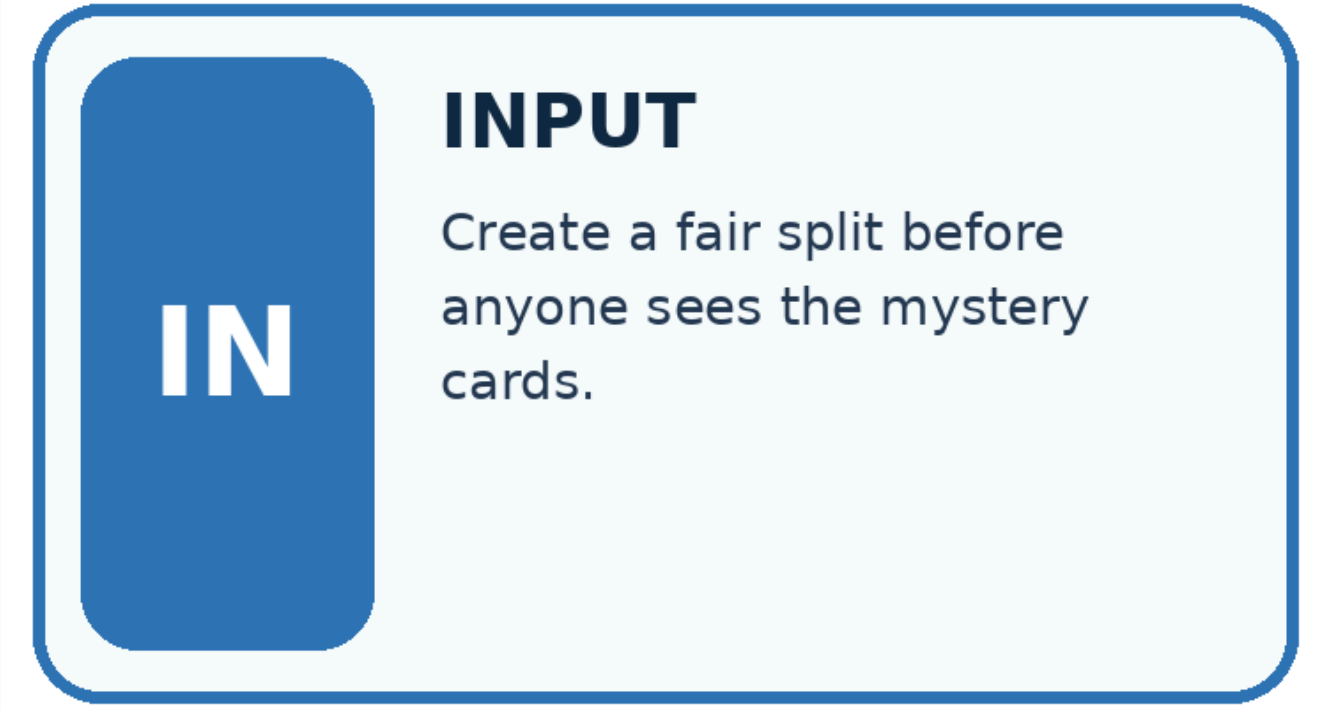
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

VISUAL **MECHANISM**

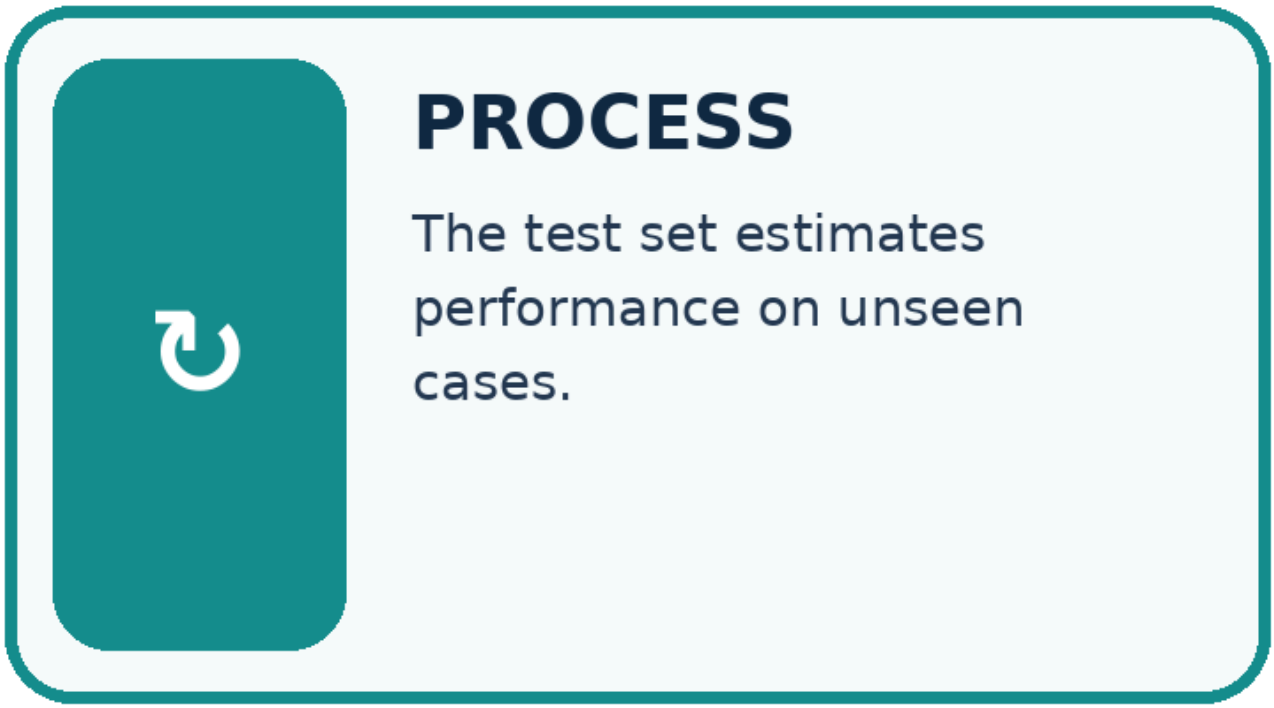
Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.



</details>

↓


PROCESS
The test set estimates performance on unseen cases.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Keep each trip in one role

WHOLE GROUPS	ROLE	PURPOSE
A B C with all frames	Training	Fit from examples
D with all frames	Validation	Compare choices
E F with all frames	Test	Final check after choices

Keep related frames together. Separate group IDs prevent this form of overlap; they do not prove that every other source of leakage is absent.

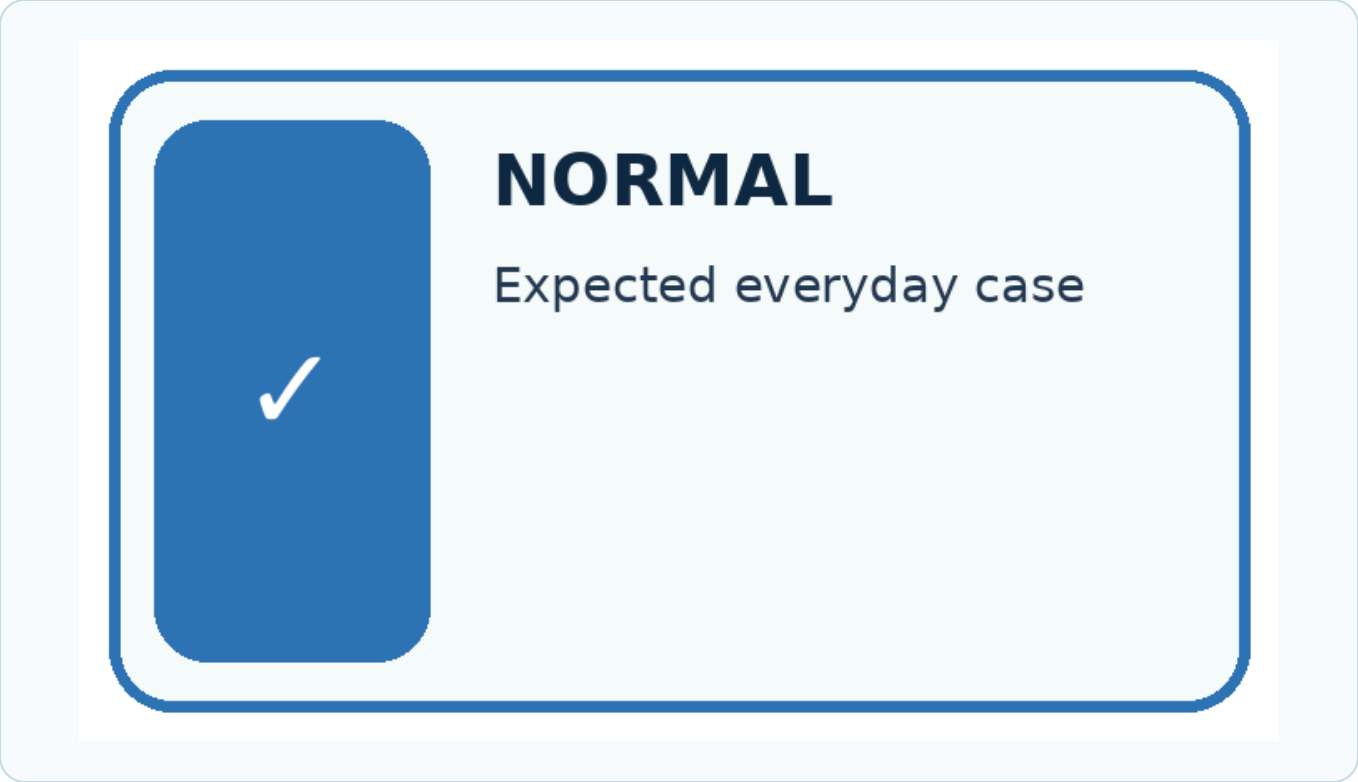
Connect the picture and the explanation

MAIN LESSON · SECTION 4

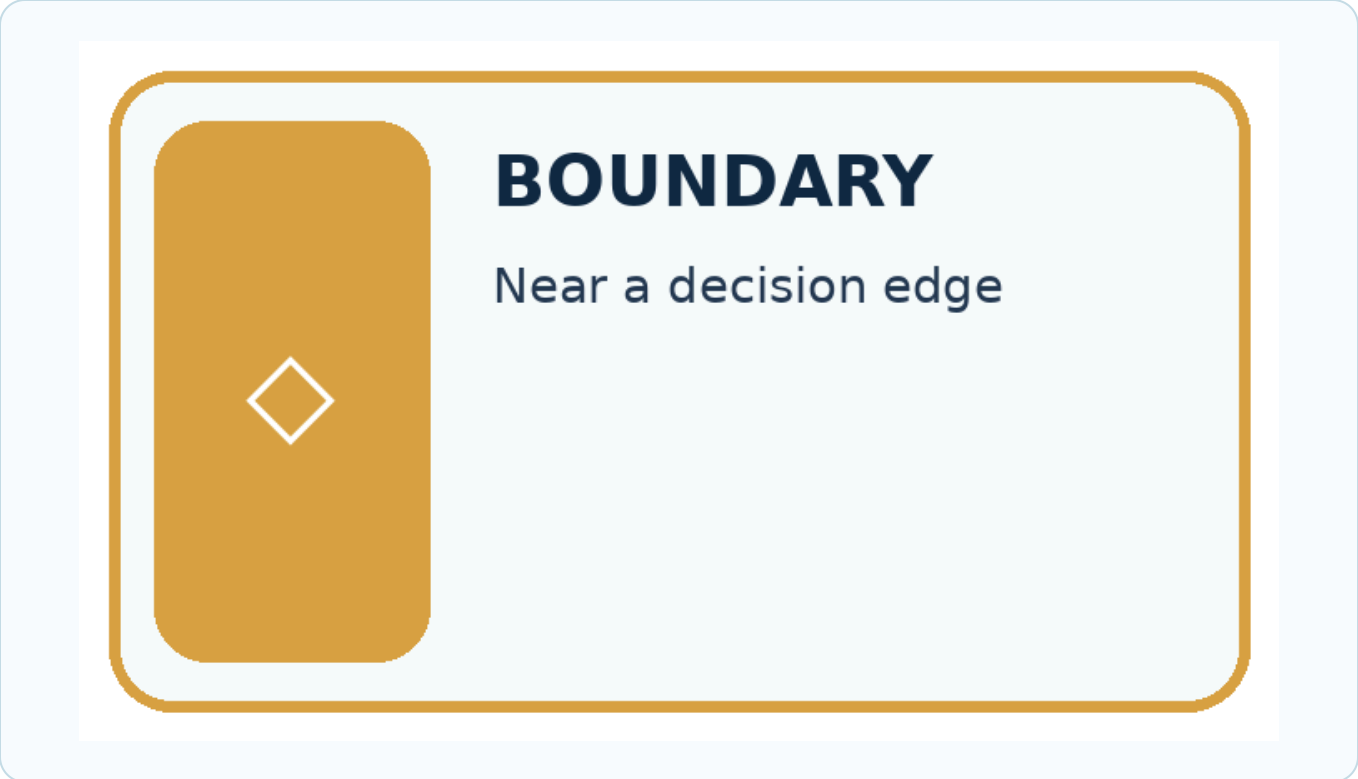
Choose a case that exposes related-record overlap

> Unique filenames do not make related examples independent.

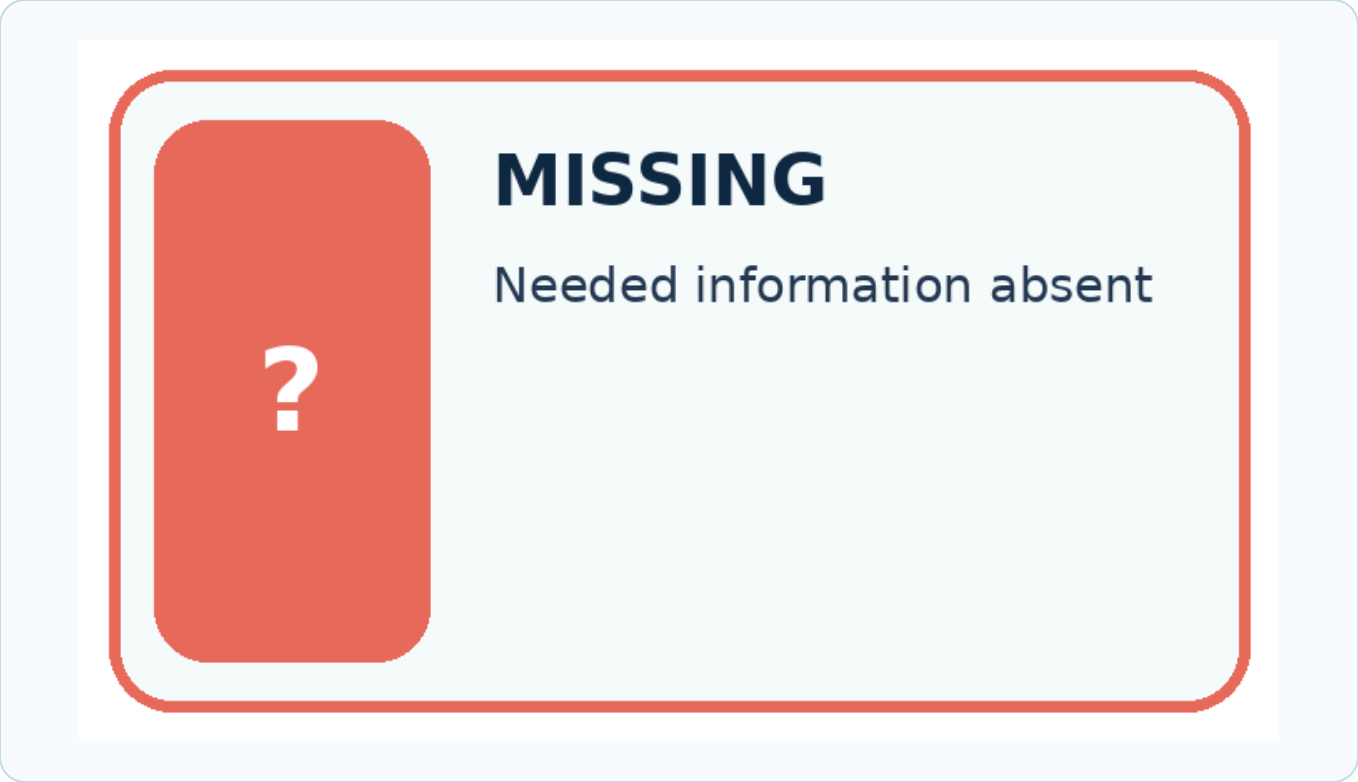
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Give A2 a new filename without changing the book it depicts.

▼ Check after predicting

Prediction: Its **group** is still A; the original related examples still cross the split.

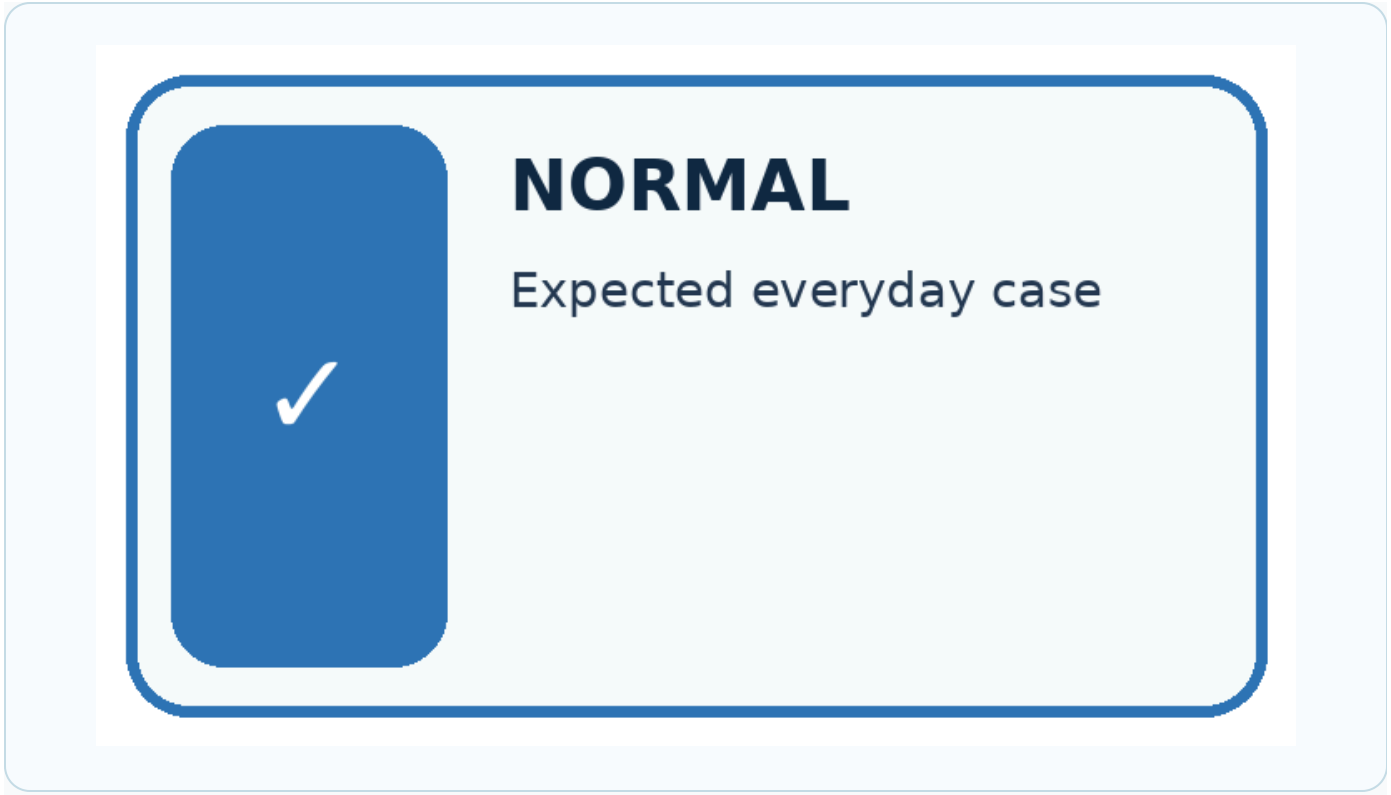
Reason: Renaming a row does not remove the relationship between scans.

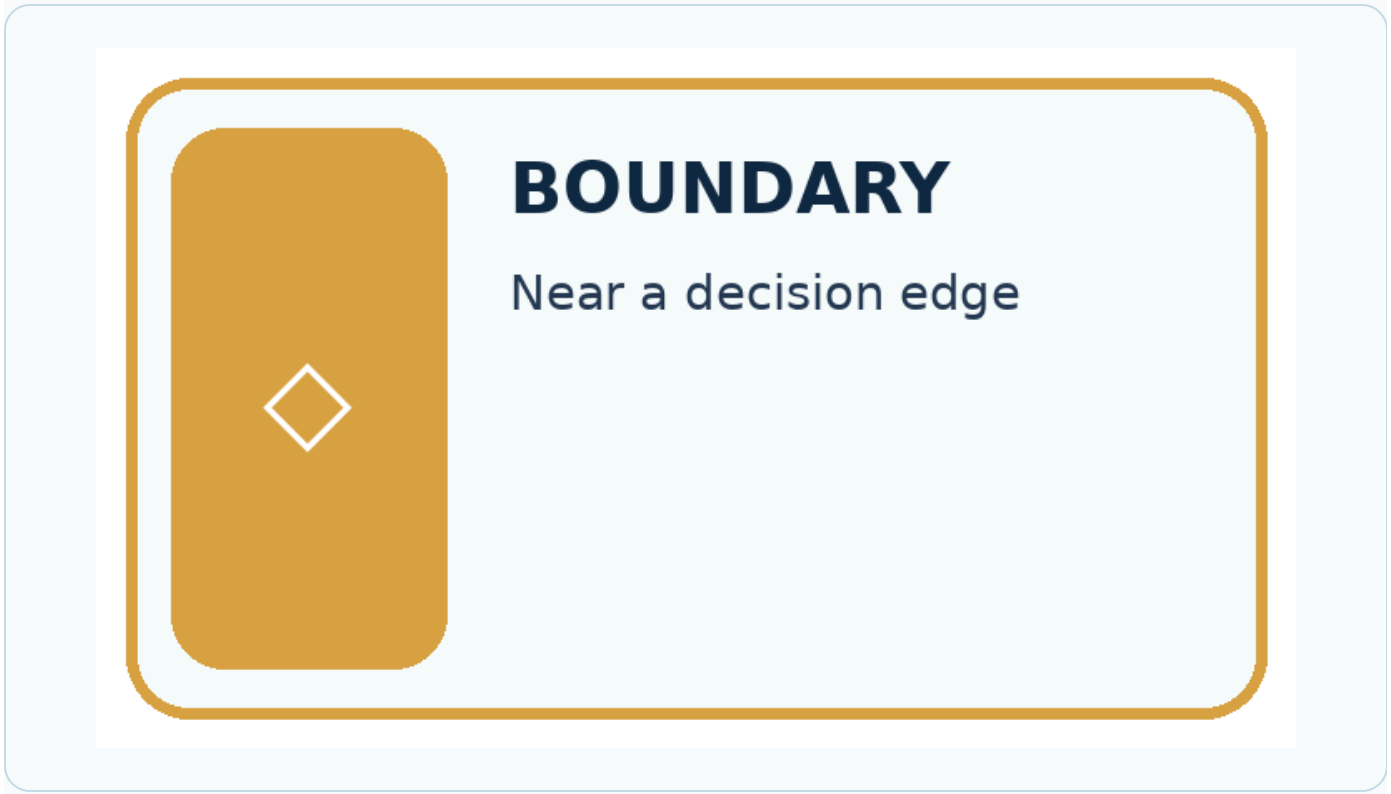
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Audit the trip plan, then a separate book-scan plan

> Trips and books are different group units for different claims.

Diagnose a trip split across roles

Suppose A1 and A2 are in **training** while A3 is in test. The frame IDs differ, but A appears in both sets. Shuffling the rows does not remove this relationship.

For our new-trip claim, move all A frames to one chosen role and record the revised assignment before evaluating a model. Do not copy A3 into both roles.

Count every frame once and record the assignment

Audit that every source frame occurs exactly once, no **group** crosses roles and the assigned counts sum to the source count. In our worked plan the row counts are six **training**, two validation and four test: twelve total.

A fixed random seed can reproduce a random split, but it cannot make a flawed **group**ing choice fair.

A split record names the **prediction** goal, source version, **group** definition, exact assignments, row and **group** counts, date and any random method or seed used. For this hand-assigned example, record that there was no random sampling. Keep label-rule versions and unresolved review statuses traceable as in Week 8.

Predict the D2-move audit

A learner moves D2 from validation to training but leaves D1 in validation. Give the new row counts and shared trip ID. Explain why different frame IDs and a fixed seed do not repair the overlap.

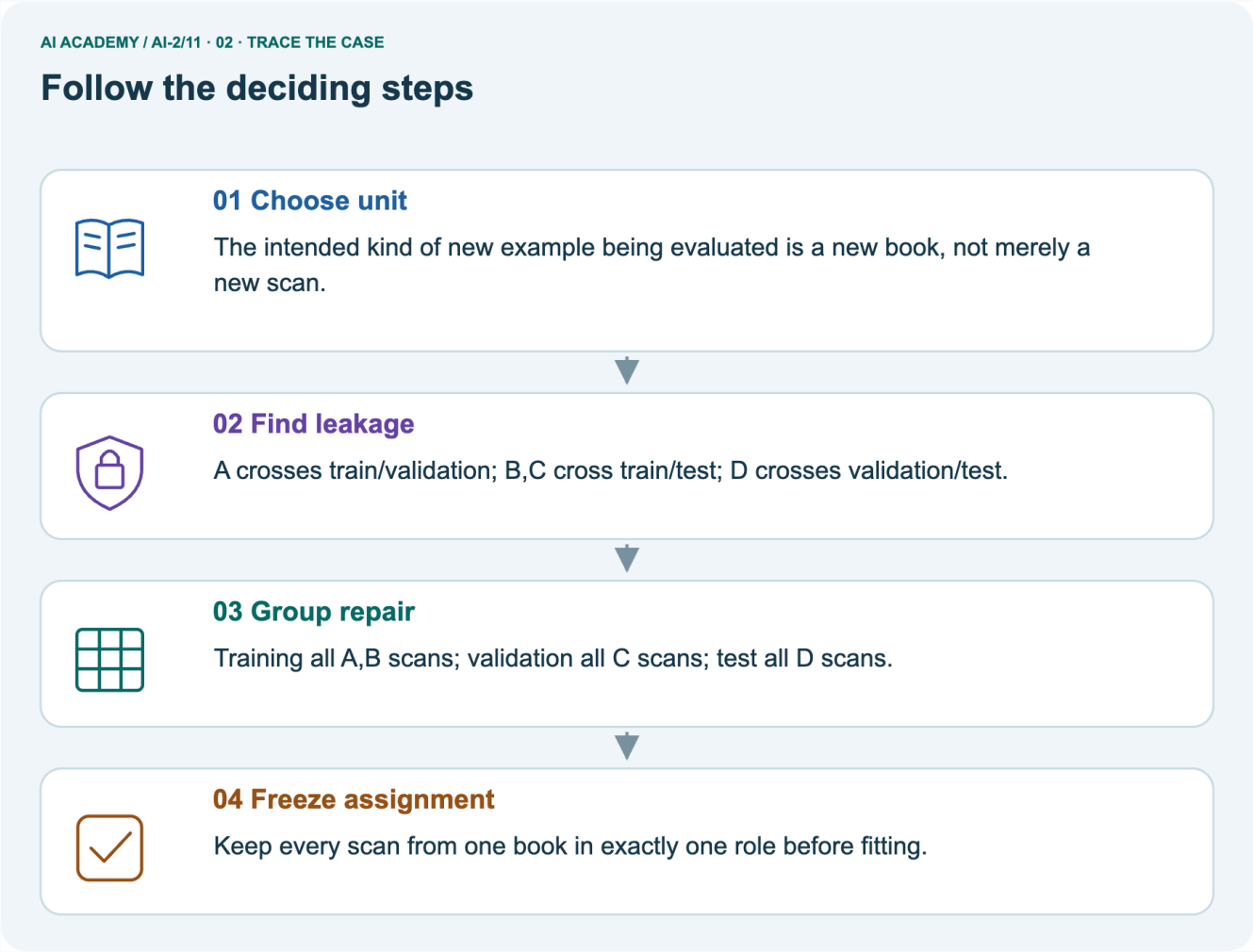
Introduce the separate new-book evaluation

A fictional catalogue model learns from scans of book covers and will face previously **unseen** books.

Book A has scans A1,A2; B has B1,B2; C has C1,C2; D has D1,D2.

Proposed **training** A1,B1,C1; validation A2,D1; test B2,C2,D2.

Trace the book-group repair



Follow the labelled picture with the matching explanation. Open the vector diagram

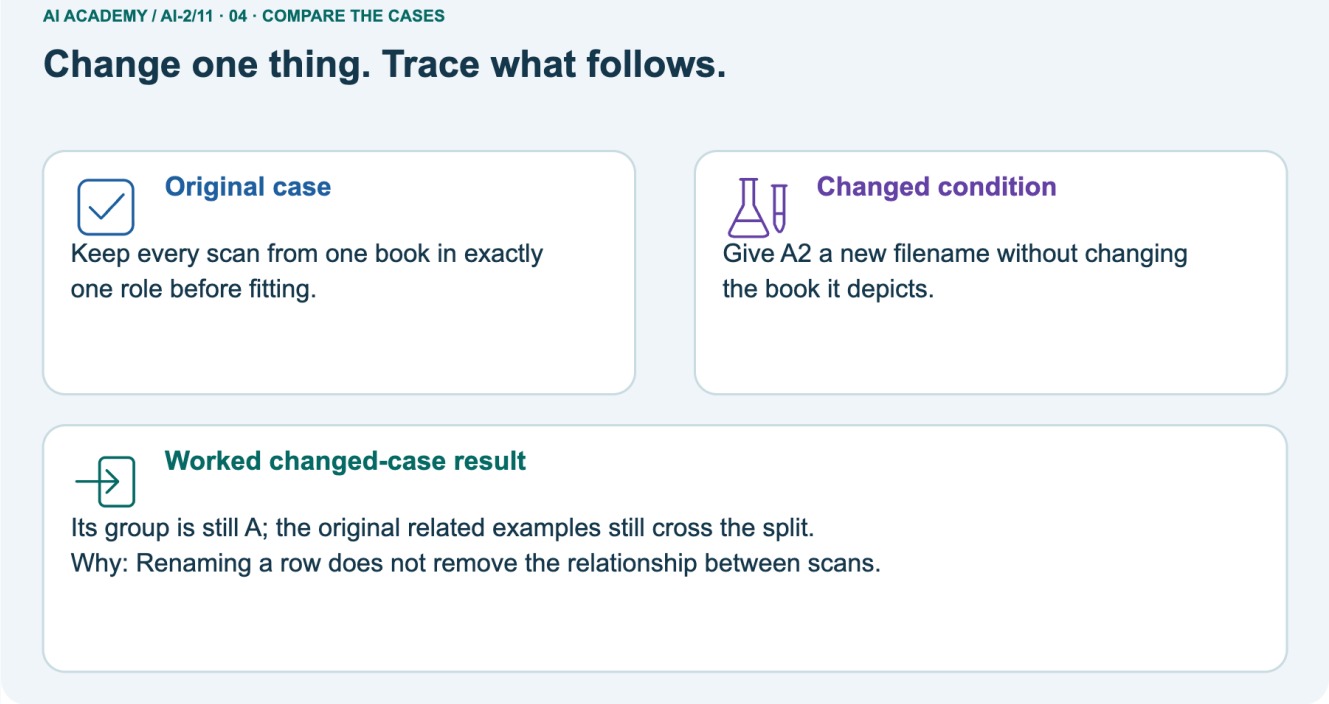
Different scan IDs do not make examples independent books. A model may recognise repeated cover details from another view. **Group**ing by book matches the intended claim about new books and makes the split inspectable.

The repaired split has four **training**, two validation and two test scans, but only two, one and one books respectively.

Predict what renaming A2 changes

- Change one thing: Give A2 a new filename without changing the book it depicts. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain book counts and limited diversity

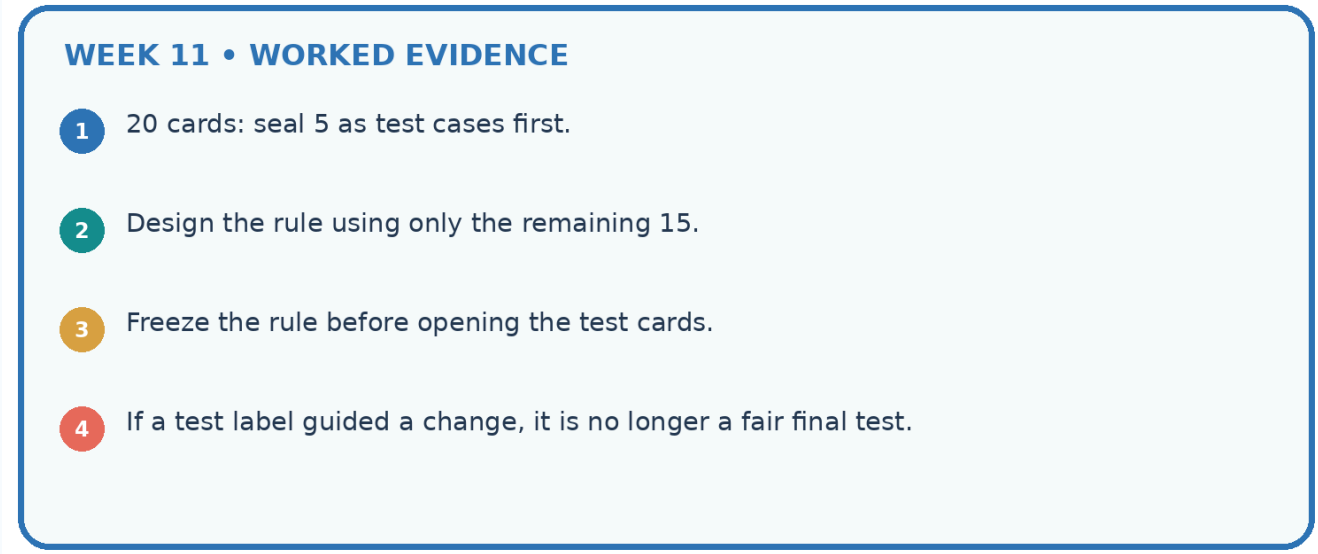
- Read the labelled picture: How many distinct books are in each repaired role, and why report that as well as scan counts?
- One validation book and one test book provide very limited diversity even though no **group**s overlap.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. 20 cards: seal 5 as test cases first.

2. Design the rule using only the remaining 15.

3. Freeze the rule before opening the test cards.

4. If a test label guided a change, it is no longer a fair final test.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

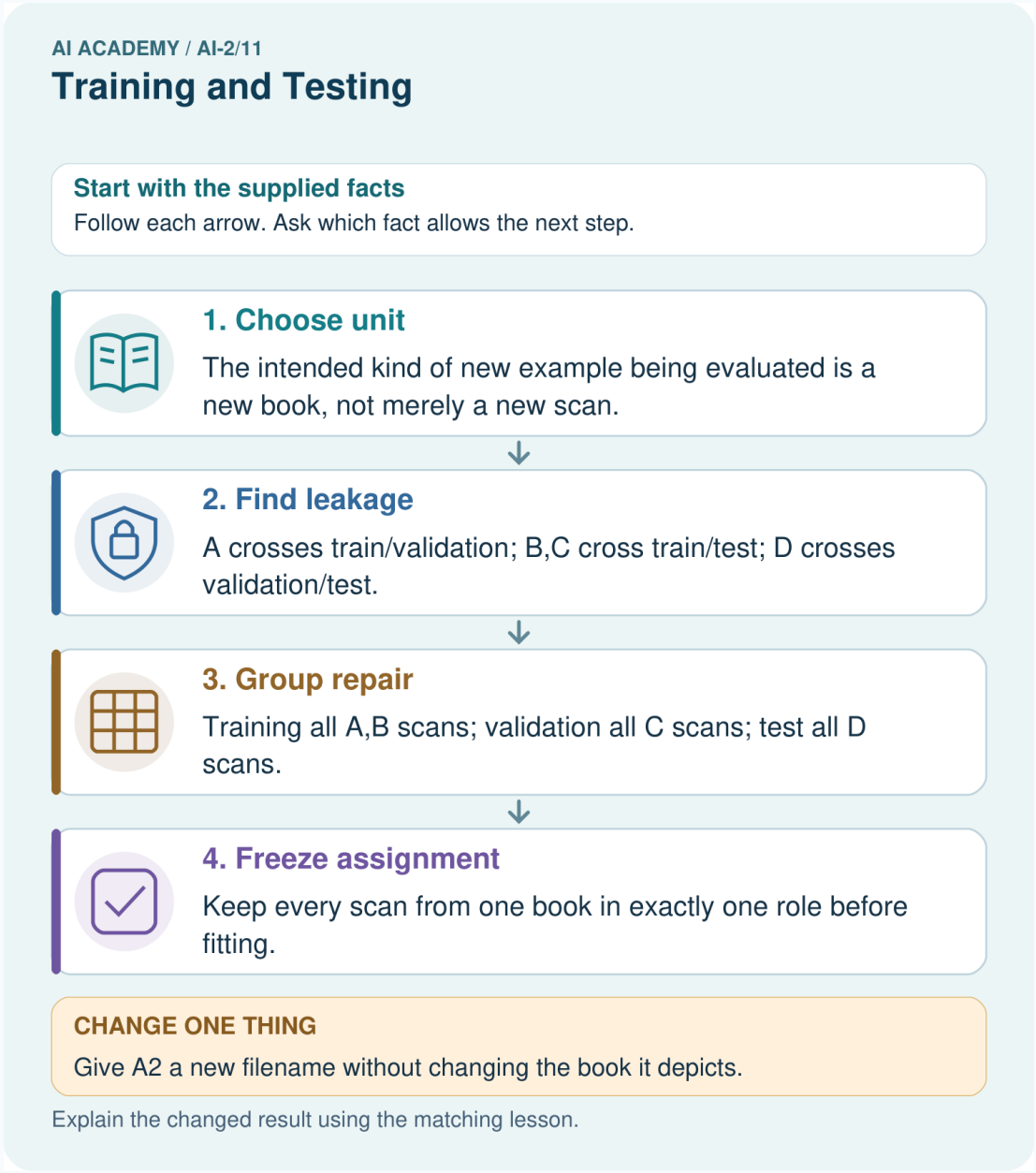
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Training and Testing

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Choose unit	The intended kind of new example being evaluated is a new book, not merely a new scan.
2. Find leakage	A crosses train/validation; B,C cross train/test; D crosses validation/test.
3. Group repair	Training all A,B scans; validation all C scans; test all D scans.
4. Freeze assignment	Keep every scan from one book in exactly one role before fitting.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

A clean group audit does not rule out every leak

> Check source records, preprocessing and the scope of the claim.

Fit learned preprocessing on training only

Fit learned pre**process**ing, such as scaling values using an average, on **training** data and apply the stored transformation to the other sets.

Using the full dataset to learn such values would let held out information influence development. We will implement pre**process**ing later; **group** separation alone does not solve that issue.

Preserve exclusions and corrections in the evaluation record

Record exclusions and label-review policies before scoring. Do not remove difficult test cases because they reduce the score. If a genuine data error requires repair, document it and its effect on the evaluation rather than claiming the old test stayed untouched. Next week, Accuracy as a Fraction, you will use explicit predictions and comparison labels to count matches; keep the split's role and scope visible when interpreting a score.

Challenge an unseen-route claim

> **All trip IDs are separate, but every trip uses the same route. Can you claim the model works on unseen routes? Explain the limitation and propose a split unit for that new claim.**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Testing on training examples proves the system will generalize.
Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Match the split unit to the intended use

> New trips, new routes and future events require different separations.

Choose the unit for the stated claim

For new trips on a familiar route, holding out whole trips addresses within-trip similarity. For new routes, hold out entire routes so all their trips stay together.

For forecasting future events, a time-respecting split may be needed: **training** on later events to predict earlier ones would not represent that intended use. The evaluation question determines the separation.

Explain why a percentage is not a fairness guarantee

There is no universal percentage that makes a split valid. Large **group**s can make row counts unequal.

Check that the sets can support the intended comparison, including enough relevant cases and labels, without moving individual related frames just to hit a percentage.

This twelve-row example is too small to justify a reliable performance claim.

Distinguish frames from groups in W2

Use workbook W2 for the connected example “Splits and related examples”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Repair the unique-row-ID misconception

- Use the worked case for Splits and related examples. Proposed explanation: “Different row IDs mean independent examples”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the separate book-count evidence

How many distinct books are in each repaired role, and why report that as well as scan counts?

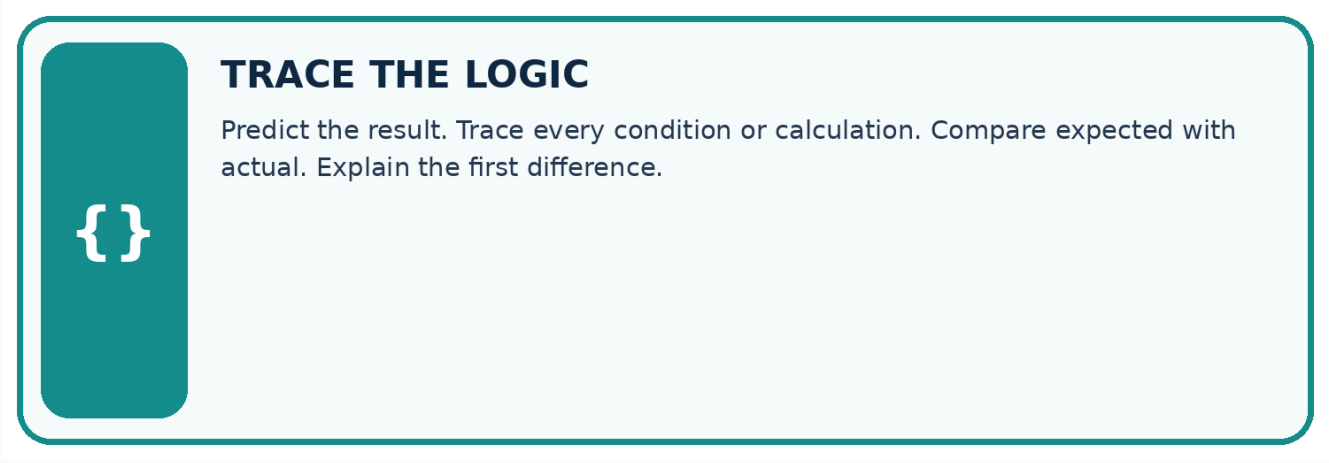
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

Practise with less help: Distinguish frames from groups

Check your reasoning

Explain the picture

> **My explanation using a pictured step and the given facts**

MAIN LESSON · SECTION 9

Audit the supplied split, a moved frame and an omitted frame

> Compare predictions with paper or Python audits while preserving each version.

The source inventory and evaluation scope

Use only the supplied twelve-frame A–F assignment in Section 2. Claim: new trips on the same route under similar conditions. Source inventory: A1,A2,A3; B1,B2; C1; D1,D2; E1,E2; F1,F2. These are fictional identifiers and group metadata; no image features, labels or model predictions are supplied. This lab audits assignment and provenance; it does not fit or score a classifier.

Predict, audit, change and repair

1. 1. Copy the source inventory and worked role assignment. Predict row counts, group counts and pairwise group overlaps before running code. 2. Verify that every inventory frame appears exactly once and its trip remains in one role. Run the supplied group-list script if Python is available; otherwise label your result a paper audit. 3. Make a separate faulty version: move D2 to training and retain D1 in validation. Derive the group sets from the actual assignments rather than reusing the old sets. Predict and record overlap and row/group counts. 4. Repair by restoring all D frames to validation, rerun the checks and preserve both versions. 5. Introduce an omitted E2 on a separate copy: explain why unchanged group sets can miss the missing row. Compare the assignment against the source inventory. Do not use independent P–S records for shared practice.

A split-specific evidence record

Audit field	Record separately for each version
Prediction goal and source version	New trips; fictional twelve-frame source
Training / validation / test	Exact frame lists, row counts and distinct trip counts
Group overlap	Training-validation; training-test; validation-test
Inventory completeness	Missing frames, repeated frames and wrong group metadata
Method and observation	Hand-assigned; no random sampling; paper audit or executed output
Repair and limit	One documented assignment change; no performance claim

Explain the revealing check

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

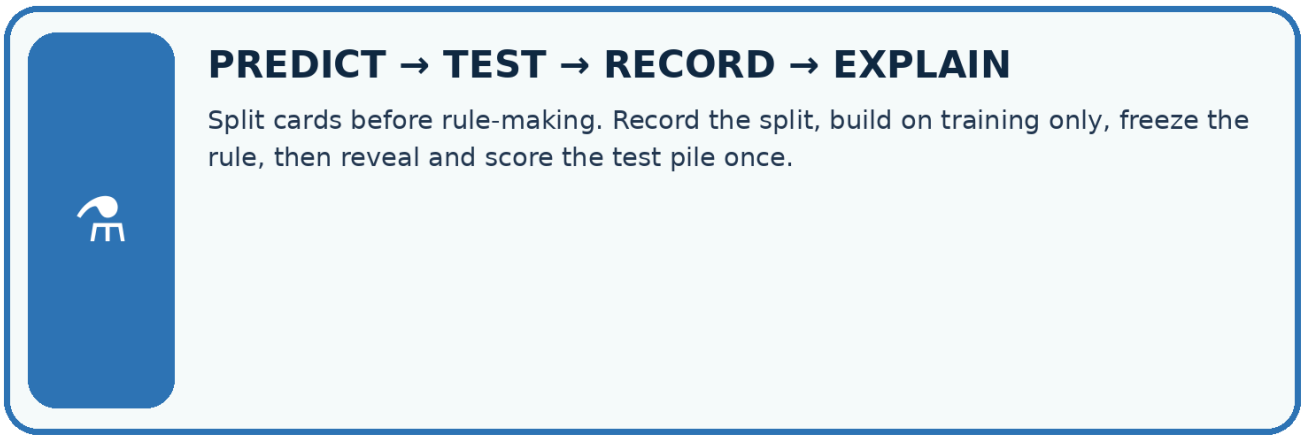
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Audit the supplied A–F frame assignments before any fitting: count rows and groups, detect a shared trip, repair the assignment and check source completeness. No model is fitted and no test accuracy is computed in this investigation.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

Plan → try → check

Try alone: audit and repair the different P–S trips

> List pairwise overlaps and every repaired frame before consulting the key.

The reserved source and proposed assignment

Eight frames come from trips P (P1 P2 P3), Q (Q1 Q2), R (R1), S (S1 S2). A proposed split is **training** P1 Q1 R1; validation P2 S1; test Q2 P3 S2. Identify shared **group**s for each pair of sets.

Repair it by assigning P and Q to **training**, R to validation and S to test. List all frames in each role and give row and **group** counts. State one limit of the repaired plan.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after the attempt

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Audit and repair independently

Week 11 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

You tune a rule using cards 1 to 8 and test it on cards 9 and 10. After seeing card 9's error you change the rule. Is card 9 still untouched test **evidence**?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Keep final user profiles hidden until weights and thresholds are fixed.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

A team tests five rule versions on the same ten books and chooses the highest score. It calls the final score an unbiased **unseen** test result. Explain the issue and revise the evaluation plan.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Design the recommender’s evaluation split before scoring

> Choose related-record groups that match the intended claim.

This week’s concrete portfolio contribution

For the fictional recommender, write the intended new-example claim first: new books, new editions, or new reader profiles are different targets. Identify related records that could cross the split, assign groups to roles, and keep frame/scan-like counts separate from distinct books or profiles. Use invented profiles. Keep final examples aside while choosing rules, weights or thresholds; this week designs that record rather than training or scoring the system.

Group the recommender's related reviews, editions or repeated-user examples according to its intended evaluation claim.

Keep the system, evidence and human choices inspectable

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

> **Write or draw your reasoning and evidence.**

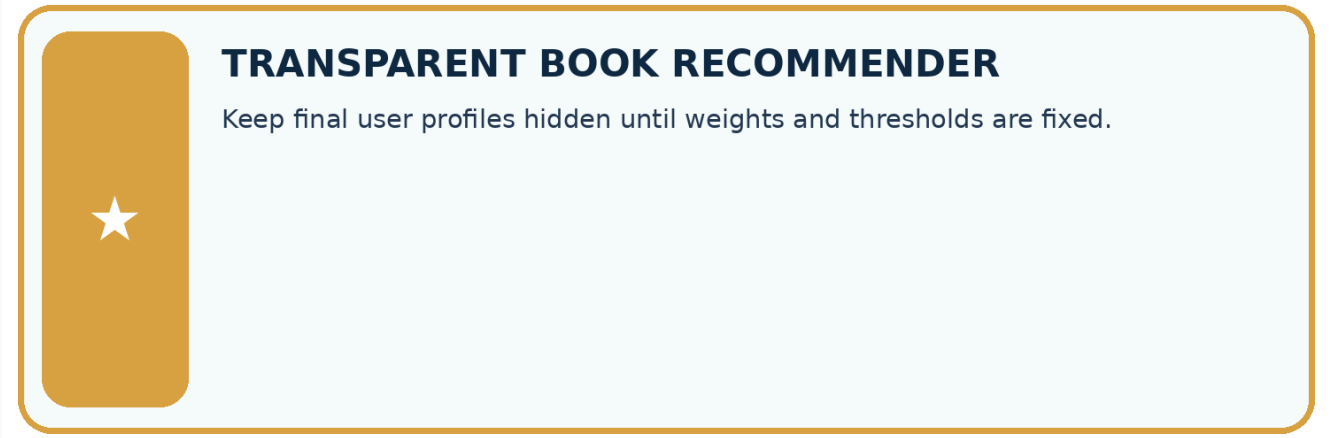
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Keep final user profiles hidden until weights and thresholds are fixed.

Evidence for your project

MAIN LESSON · SECTION 16

Finish: justify the separation using the claim

> State both the audit result and a remaining limitation.

Explain without notes

- Close your notes. Explain this goal in your own words: Choose a split unit that matches the intended **prediction** claim

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
training set	Examples used to adjust a model's parameters or fit learned preprocessing.	
test set	Data reserved for evaluating a selected system; final test data must not influence its development.	
unseen	Not previously used for the development, practice or selection being evaluated.	
leakage	Use of information during development or prediction that should be unavailable for a fair evaluation of the intended task.	

Terms for the connected explanation

Word	Meaning
Validation set	Separate examples used to compare choices during development

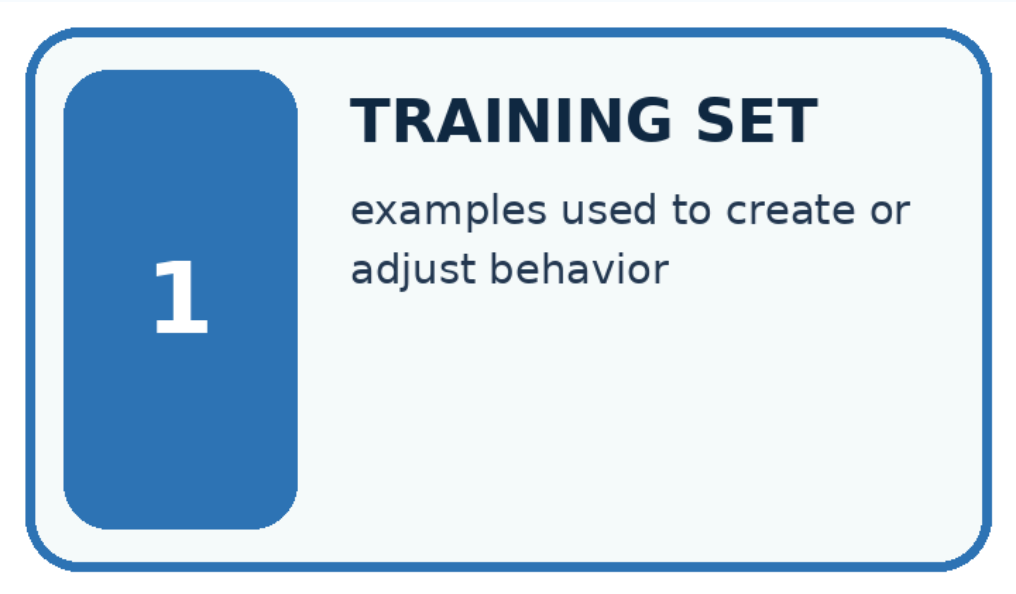
Word	Meaning
Group	Related examples that must stay together for a stated evaluation goal

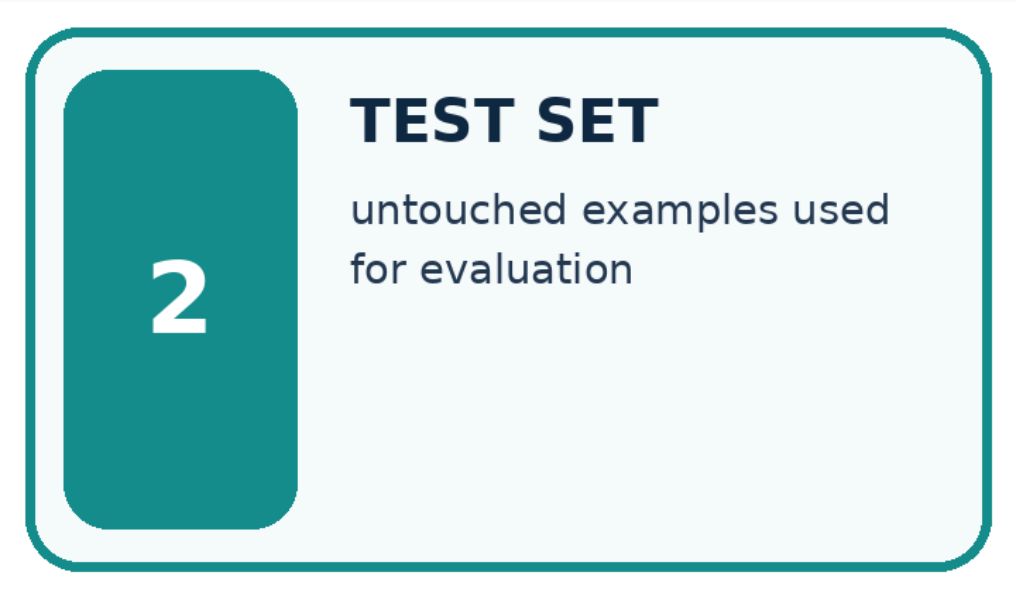
▼ Earlier wording, alternatives and picture transcripts

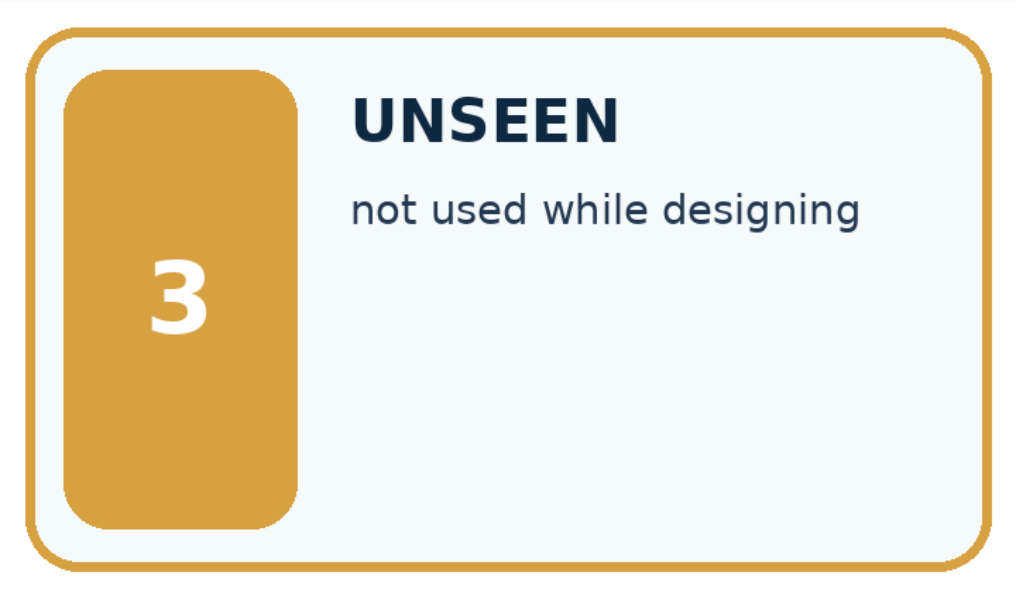
VOCABULARY LAB

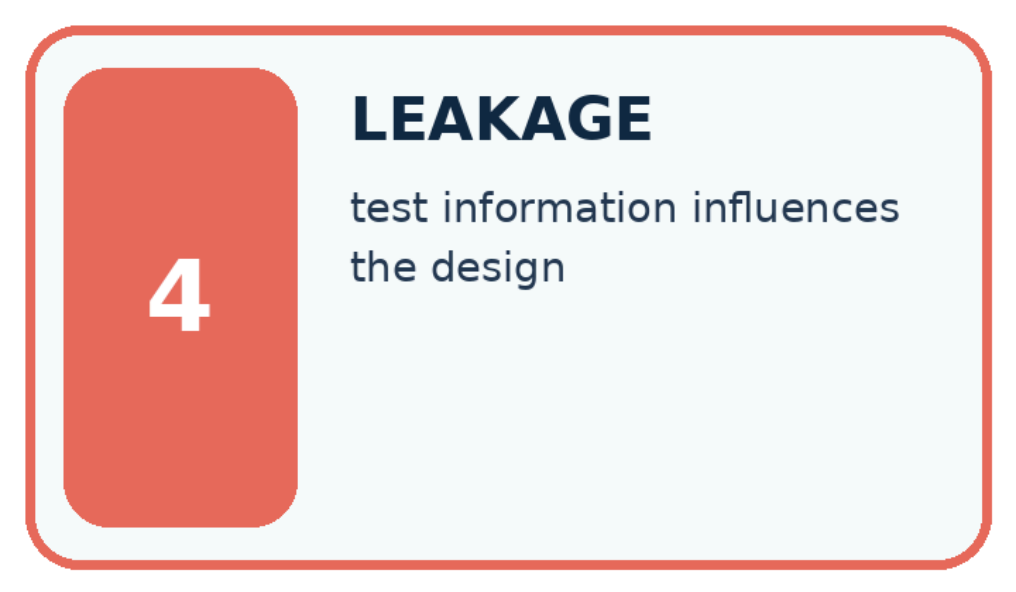
Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
TEST SET
untouched examples used for evaluation


3
UNSEEN
not used while designing



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

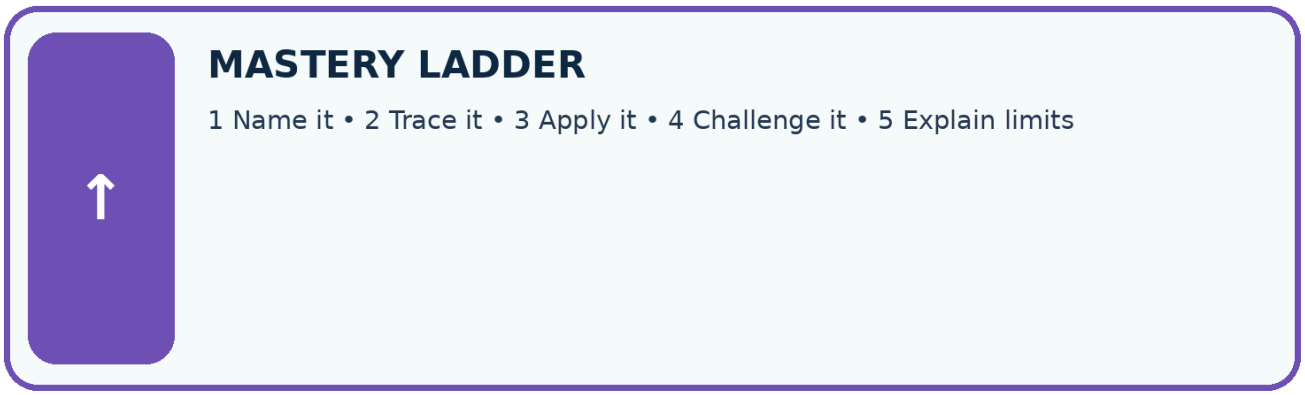
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Audit and repair independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

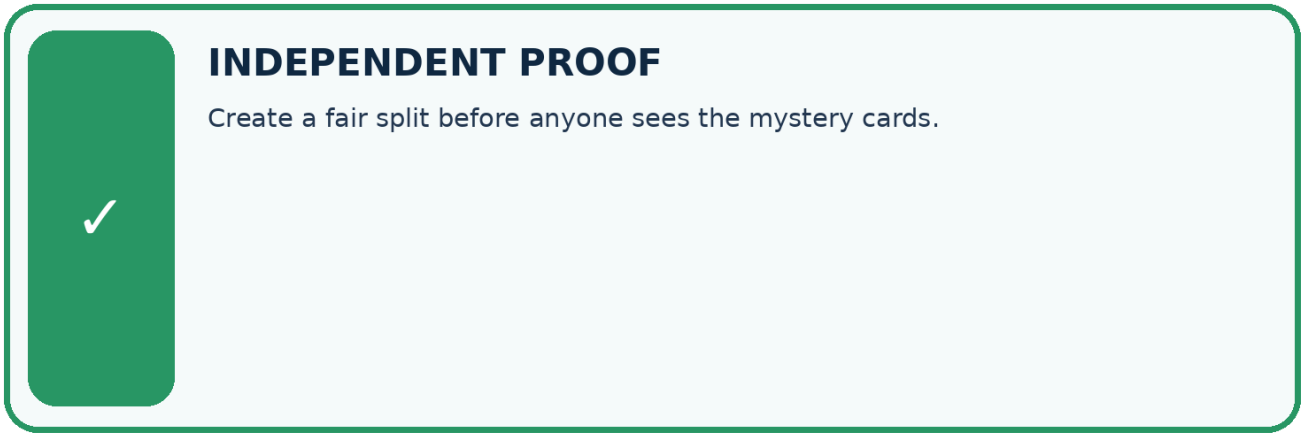
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Audit and repair independently is the selected independent case.



EXPLANATION CHECK State the prediction claim, keep related records together, count groups separately and limit the conclusion.

Write or draw your reasoning and evidence.

▼ **Extra practice: Separate development from fresh evaluation**

REFERENCE · SECTION 13

Extra practice: Separate development from fresh evaluation

> Try the question before opening its worked explanation.

Read the supplied case and question

A rule is developed and evaluated using the same books A–H.

Why does this not establish **unseen** performance?

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Those cases influenced the rule. Reserve different labelled books that do not influence construction or revision.

Evidence or stage	Value or result
Books A–H	Influenced the rule
Reserved books	Did not influence construction
Revision	Keep final cases separate

▼ Earlier wording, alternatives and picture transcripts

Week 11 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Separate development from fresh evaluation

A rule is developed and evaluated using the same books A–H.

Why does this not establish **unseen** performance?

Those cases influenced the rule. Reserve different labelled books that do not influence construction or revision.

▼ **Reference: original lesson checklist**

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Choose a split unit that matches the intended prediction claim	Explain the deciding step, show a trace or test, and state one limit.
Keep related examples together and audit group overlap	Explain the deciding step, show a trace or test, and state one limit.
Record split roles counts and limits before model evaluation	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional catalogue model learns from scans of book covers and will face previously **unseen** books.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional catalogue model learns from scans of book covers and will face previously **unseen** books.
- Which labelled step matches this detail? Keep every scan from one book in exactly one role before **fitting**.
- 1. **Group** repair
- 2. Choose unit
- 3. Find **leakage**

- 4. Freeze assignment

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Freeze assignment Keep every scan from one book in exactly one role before **fitting**. Different scan IDs do not make examples independent books.

A model may recognise repeated cover details from another view. **Group**ing by book matches the intended claim about new books and makes the split inspectable.

The repaired split has four **training**, two validation and two test scans, but only two, one and one books respectively.

Changed-case question

- Changed-case practice: Give A2 a new filename without changing the book it depicts. Which conclusion is supported by the supplied facts?

- 1. Its **group** is still A; the original related examples still cross the split.

- 2. This one example guarantees correct results for every future **input**.

- 3. We can decide the result without examining the changed **input** or the procedure.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Its **group** is still A; the original related examples still cross the split. Renaming a row does not remove the relationship between scans.

Repair and save your work

Review Human Classifier Simulation (ai-2/10). Goal: Construct a classifier with a stated **input** format, selected rule and recorded **output**.

Short practice: Why are J and M initially treated differently even though both are illustrated mysteries?. Return to this week and explain what you changed.

Review Operational Labels (ai-2/8). Goal: Apply a category definition using explicit **evidence** and boundary rules. Short practice: Why can we not resolve the disagreement merely by taking a majority of reviewers?.

Return to this week and explain what you changed.

Review Loops and Stopping (ai-2/5). Goal: Trace variable values before and after each loop iteration. Short practice: Why is report 0 correct for the original loop, and why is equality-only stopping risky here?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work
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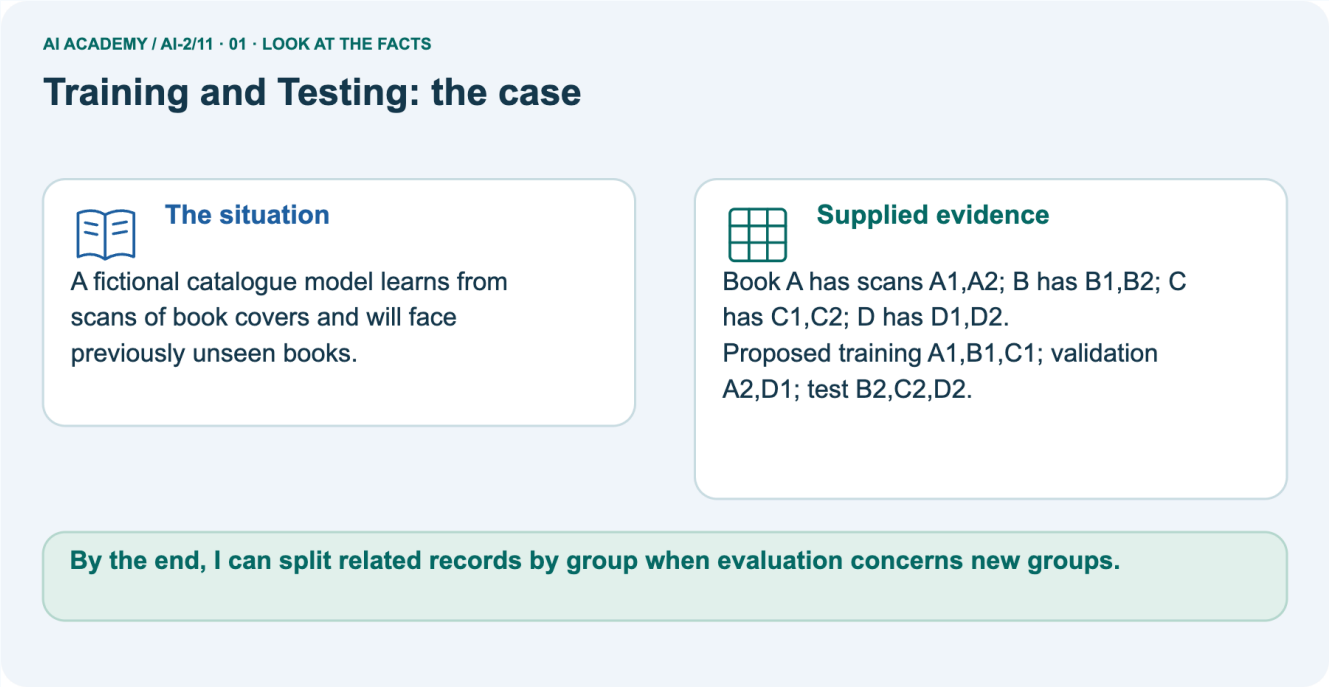
<details>
<summary>Reference: the six-picture journey</summary>
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REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Training and Testing: the case



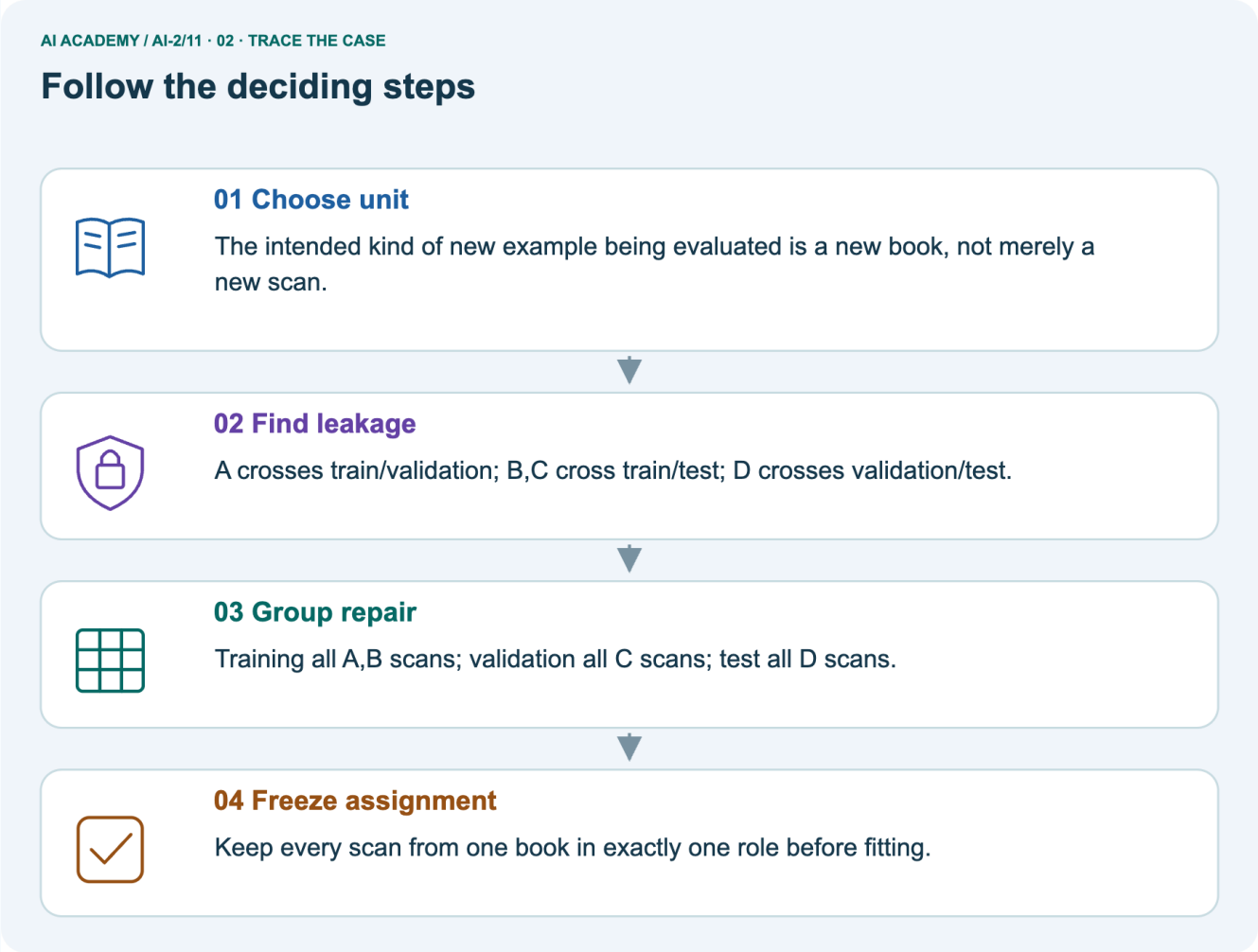
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional catalogue model learns from scans of book covers and will face previously **unseen** books. Supplied facts: Book A has scans A1,A2; B has B1,B2; C has C1,C2; D has D1,D2.

Proposed **training** A1,B1,C1; validation A2,D1; test B2,C2,D2.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

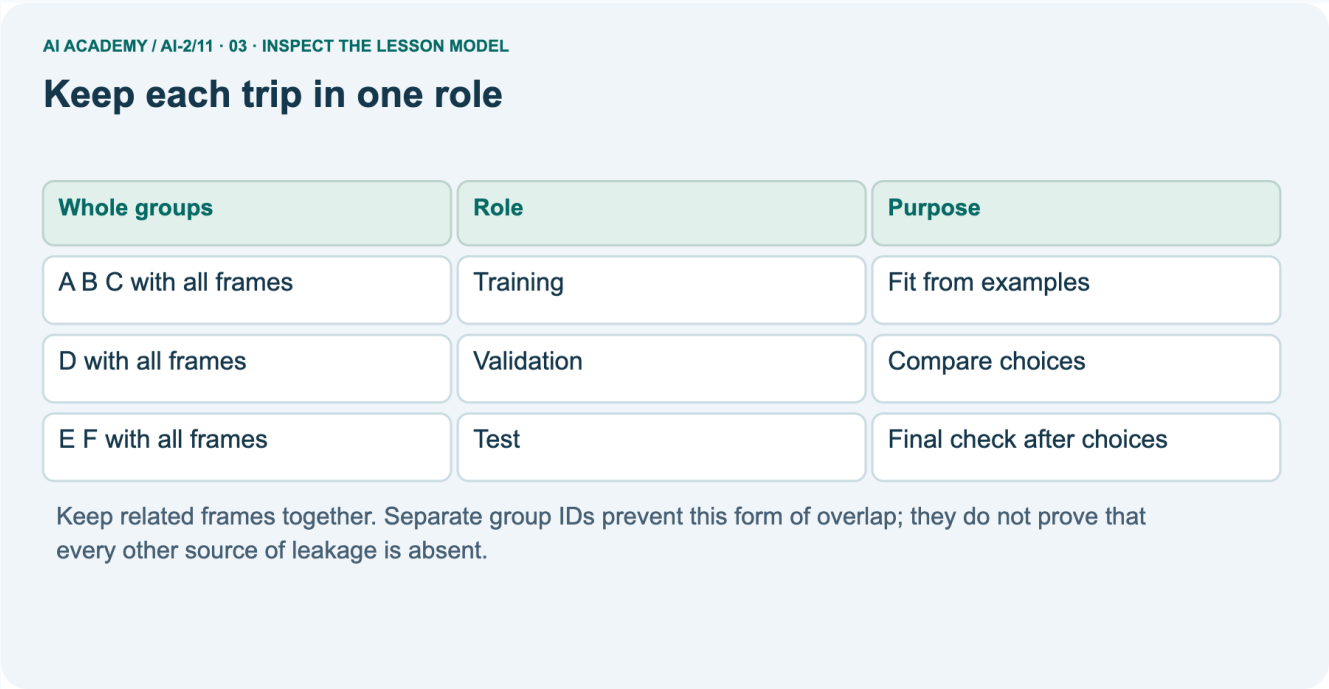


Follow the labelled picture with the matching explanation. Open the vector diagram

Choose unit: The intended kind of new example being evaluated is a new book, not merely a new scan. → Find **leakage**: A crosses train/validation; B,C cross train/test; D crosses validation/test. → **Group** repair: **Training** all A,B scans; validation all C scans; test all D scans. → Freeze assignment: Keep every scan from one book in exactly one role before **fitting**.

Point to each numbered stage and explain which recorded **input** it uses.

03. Keep each trip in one role

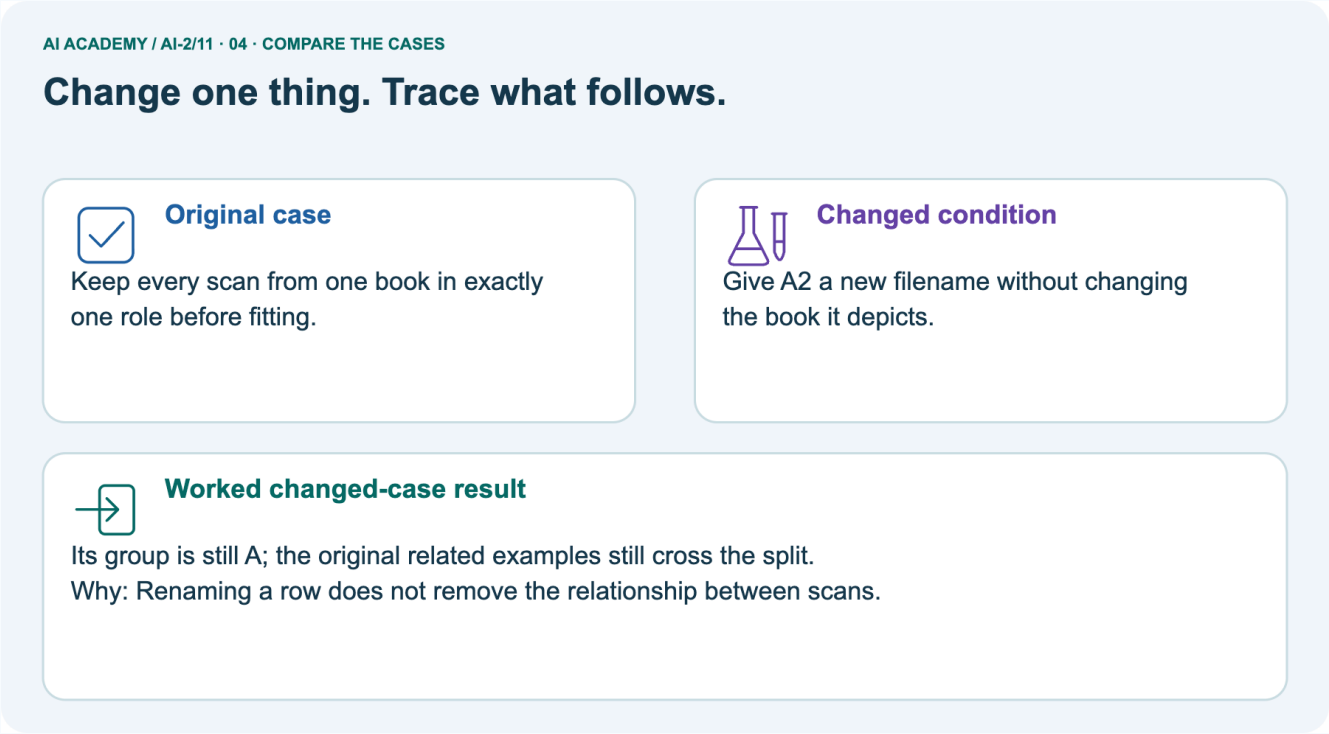


Follow the labelled picture with the matching explanation. Open the vector diagram

Keep each trip in one role. A B C with all frames | **Training** | Fit from examples. D with all frames | Validation | Compare choices. E F with all frames | Test | Final check after choices

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



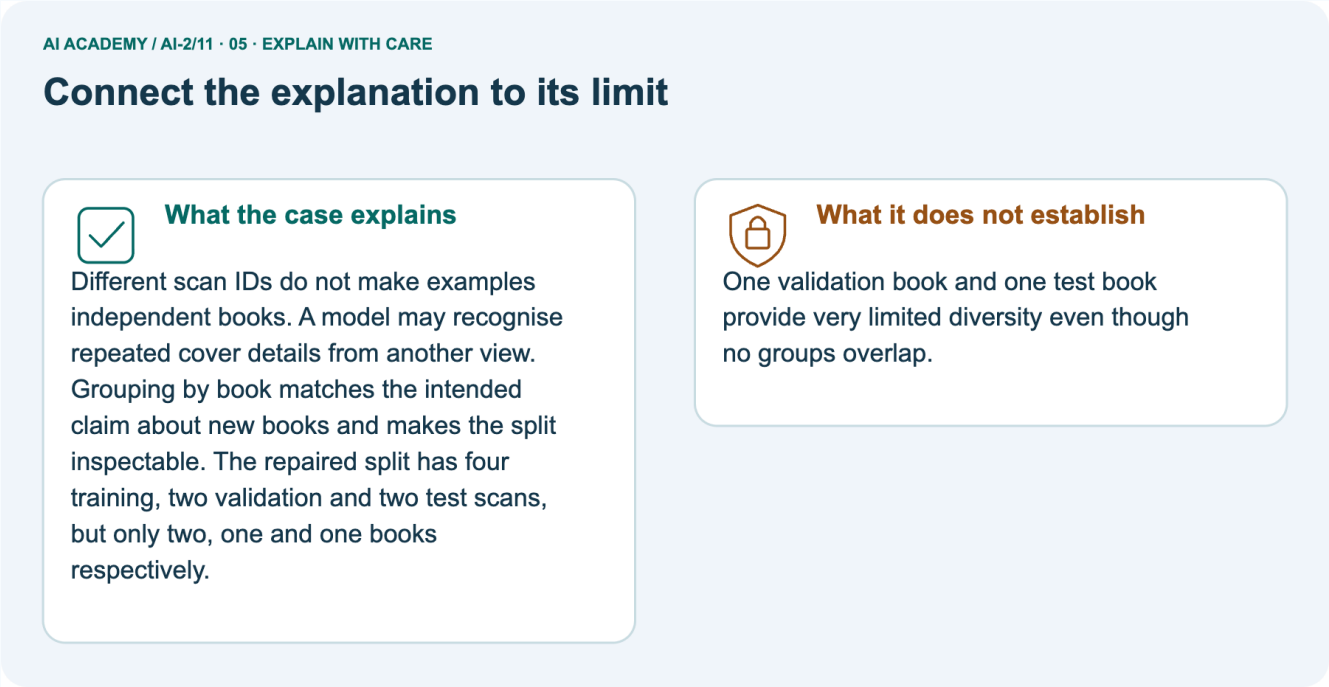
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Keep every scan from one book in exactly one role before **fitting**. Changed condition: Give A2 a new filename without changing the book it depicts.

Expected result: Its **group** is still A; the original related examples still cross the split. Reason: Renaming a row does not remove the relationship between scans.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



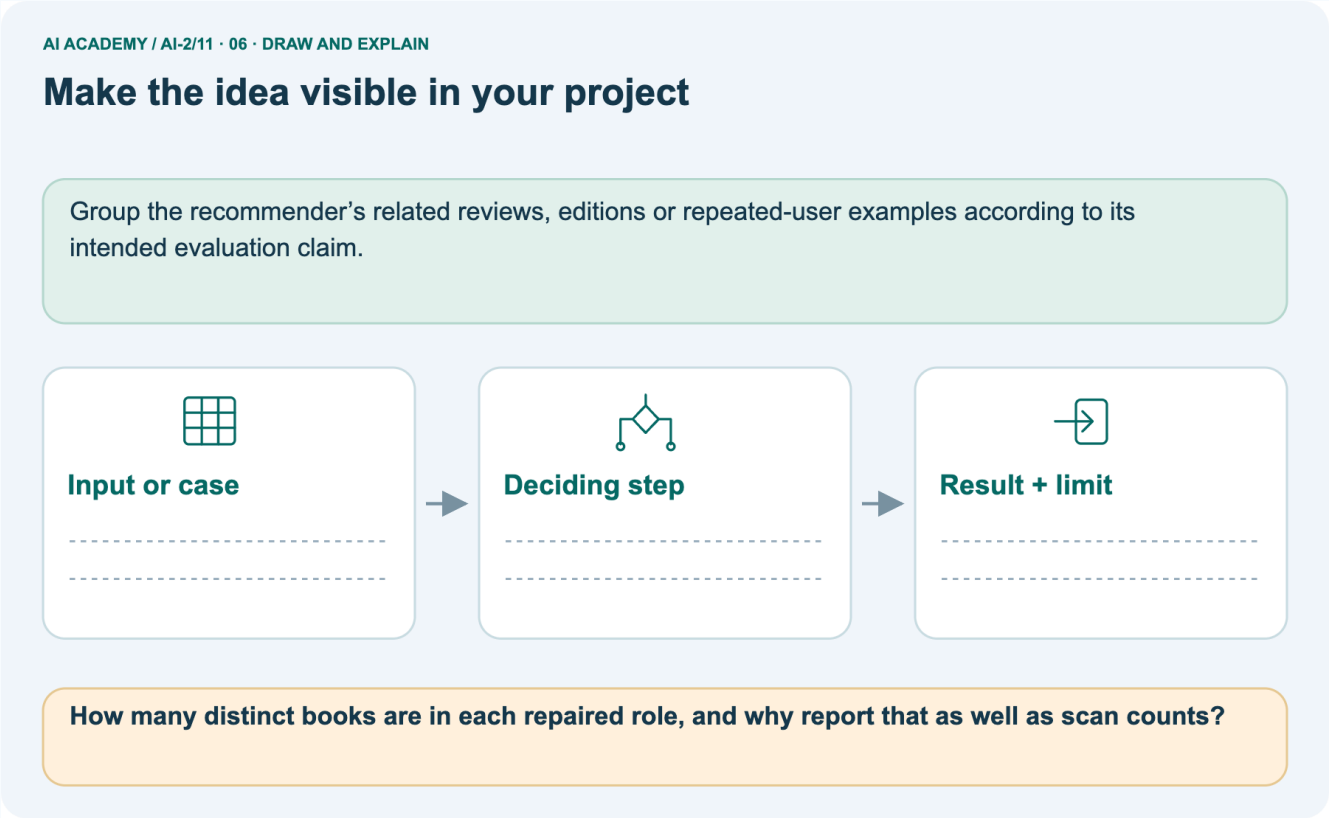
Follow the labelled picture with the matching explanation. Open the vector diagram

Different scan IDs do not make examples independent books. A model may recognise repeated cover details from another view. **Group**ing by book matches the intended claim about new books and makes the split inspectable.

The repaired split has four **training**, two validation and two test scans, but only two, one and one books respectively. Limit: One validation book and one test book provide very limited diversity even though no **group**s overlap.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Group the recommender's related reviews, editions or repeated-user examples according to its intended evaluation claim. Draw your **input**, deciding step, result and limit.

Supported question: How many distinct books are in each repaired role, and why report that as well as scan counts?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Training and Testing: the case

02. Follow the deciding steps

03. Keep each trip in one role

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.